Sound and communication

Sound is a form of energy that requires a medium for propagation, ie it can travel through a gas, liquid or solid, but not a vacuum. Vibrating objects produce sound. How quickly an object vibrates is called its frequency and is measured in cycles per second or hertz (Hz).

High notes have high frequencies and are produced by rapidly vibrating objects. Low notes have low frequencies and are produced by slowly vibrating objects. Pitch is the property of sound that we recognise as lower and higher notes.

The wavelength of a sound wave is related to frequency (and pitch). The higher the frequency, the shorter the wavelength. Wavelength is measured in metres.

1 Write these frequencies in words and numbers: 20 Hz 10 KHz

2 Give your own definition of *pitch*.

3 Using diagrams, show the relationship between wavelength and frequency for a low frequency sound and a high frequency sound.

The larynx, lungs, tongue and lips all play a part in how humans produce intelligible (understandable) sounds. The larynx ('voice box') is a hollow chamber near the top of the trachea ('windpipe') and contains the vocal cords. The vocal cords are two pairs of folded mucous membranes. The larynx is supported by cartilage (gristle) in places. Air passing from the lungs can cause the vocal cords to vibrate. The frequency of the sound is controlled by muscles that change the length and tension of the vocal cords. An analogy (comparison) can be made with the strings of a musical instrument that are plucked.

4 Give another name for the *voice box*.

5 Give another name for the *windpipe*.

6 Give another name for *gristle*.

7 Identify the following organ: *'The modified upper part of the trachea of air-breathing vertebrates that contains the vocal cords.'*

Animals

Apart from humans, many animals use sound to communicate. Male frogs can be very vocal. All frogs have a larynx that vibrates when air is forced through from the lungs. Male frogs of certain species have the lower part of the mouth as a large vocal sac (elastic pouch) used as a resonator to amplify their call. The sac inflates and vibrates like a drum. In some areas where frog population is high during the mating season, the noise produced by frogs can be in the region of 80 dB – about the same as a lawn mower.

8 Give three reasons why animals might communicate.

9 Give the full name for the unit *dB*.

10 Very high frequencies are used in 'echolocation' by some predators to find prey. Give an example of an animal that does this.

11 Very low sounds travel great distances under water and can be used by marine mammals to communicate. Give an example of an animal that does this.

The syrinx is the vocal organ (song box) of birds and is a special modification of the lower part of the trachea and/or bronchi.

Decide whether the following statements are true or false:

12 Humans are the only animals that use sound to communicate. _____

13 Vocal cords in humans are inflexible structures. _____

14 All sounds are produced by vibrations. _____

15 Birds generally use a vocal sac to make sounds. _____

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Fish

Fish generate sounds with their swim bladders. Many fish have muscles that insert into the swim bladder. When the muscles are contracted, the swim bladder vibrates, producing a hollow sound. Some species, like the bull Atlantic Croaker (*Micropogonia undulatus*) make sounds at feeding times, while others make noise during reproductive periods.

Fish have good hearing but do not have external ears; sound waves are conducted by bones in the skull to the sensory chambers of the inner ear. Each chamber is lined with sensory hair cells. In some fishes, the inner ear is connected to the swim bladder that can resonate and 'amplify' sound. An external ear structure isn't needed because the density of fish bodies is similar to water, which means sound waves can pass through their bodies to the inner ear. Fish also have lateral lines. These sensory organs can detect low frequency vibrations and are used by the animal to swim in schools. Sound in water travels 4.5 times faster than in air and is felt as a pressure sensation in the sensitive lateral organs on fishes' sides as pressure waves.

Insects

Insects do not use voices to make sounds; often they use special structures on their bodies, wings or legs. In crickets, only the adult males chirp by scraping front wings together around 4000 times a second, to produce their musical noise. They do this to stake out territory or to attract mates.

Adult male cicadas produce a loud song using specialised sound-producing organs called *tymbals* (shaped like a kettle drum). Each species has a specific mate-attracting song. Female cicadas do not have tymbals, but in some species the females produce clicking or snapping sounds with their wings.

Insects don't have ears in the normal sense of the word. Insects can have 'ears' on their legs (frequently just below the knee), antennae or on the body. Many insects rely heavily on their hearing. Bees and caterpillars have sensitive hairs that detect vibrations, rather than sounds travelling through the air. Some theories suggest that ears of modern insects have evolved from vibration detectors.

Grasshoppers have six pairs of chordotonal organs (primitive stretch receptor-like hearing organs), along the sides of the body. Five sets of receptors lie along the side of the abdomen, and another more complex pair lies just above the hind legs. Some grasshoppers can hear calls (under certain conditions) from up to 2 kilometres away. A male mosquito can detect the 'buzz' of a female in flight by using antennae to pick up vibrations.

Ants don't seem to be sensitive to sounds travelling through the air, but can 'hear' with their knees, picking up vibrations. Many insects detect sound waves in air by using hairlike sense organs called trichoid sensilla, rather like the sound sensors in our ears. Some researchers claim that it is likely that ants could do the same, as their antennae are covered with such hairs.

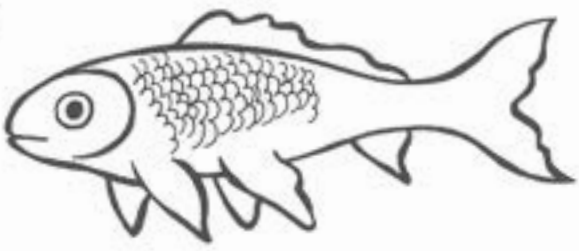
The sensitivity of those insects that can hear (butterflies, cicadas, crickets, flies and moths) is thought to extend into ultrasonic range, ie above the hearing range of humans, 20 kHz and above.

16 Draw up a table that shows all of the organisms mentioned in the information above, how they make sound and how they receive sounds.

17 What is meant by *ultrasonic*?

18 Why do fish not need external ears?

19 Does sound travel faster or slower in water compared to air?



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